MongoDB Lab – Day 4 (Aggregation)

Schema

{ roll_no, name, department, semester, city, marks }

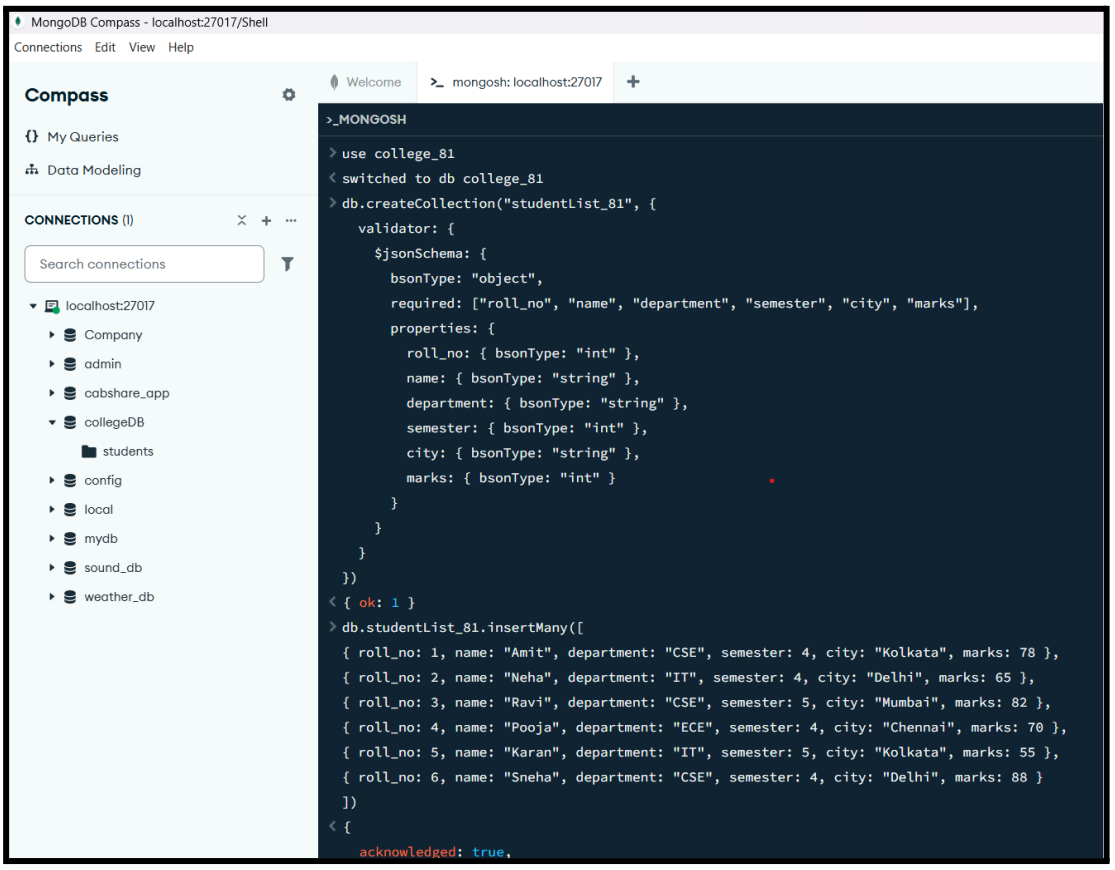
Initial Data

Roll No	Name	Department	Semester	City	Marks
1	Amit	CSE	4	Kolkata	78
2	Neha	IT	4	Delhi	65
3	Ravi	CSE	5	Mumbai	82
4	Pooja	ECE	4	Chennai	70
5	Karan	IT	5	Kolkata	55
6	Sneha	CSE	4	Delhi	88

Database Name: college_81
Collection Name: studentList_81

<u>Lab Questions</u>

Q1. Display all student records using aggregation.



MongoDB Compass - localhost:27017/Shell

Connections Edit View Help

Compass

My Queries

Data Modeling

CONNECTIONS (1)

Search connections

- localhost:27017
  - Company
  - admin
  - cabshare_app
  - collegeDB
    - students
  - config
  - local
  - mydb
  - sound_db
  - weather_db

Welcome | mongosh: localhost:27017

>_MONGOSH

```
    acknowledged: true,
    insertedIds: {
      '0': ObjectId('699b505857cb5b6c2a78cc73'),
      '1': ObjectId('699b505857cb5b6c2a78cc74'),
      '2': ObjectId('699b505857cb5b6c2a78cc75'),
      '3': ObjectId('699b505857cb5b6c2a78cc76'),
      '4': ObjectId('699b505857cb5b6c2a78cc77'),
      '5': ObjectId('699b505857cb5b6c2a78cc78')
    }
  }
> db.studentList_81.aggregate([
    { $match: {} }
  ])
< {
    _id: ObjectId('699b505857cb5b6c2a78cc73'),
    roll_no: 1,
    name: 'Amit',
    department: 'CSE',
    semester: 4,
    city: 'Kolkata',
    marks: 78
  }
  {
    _id: ObjectId('699b505857cb5b6c2a78cc74'),
    roll_no: 2,
    name: 'Neha',
    department: 'IT',
    semester: 4,
    city: 'Delhi',
```

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  - weather_db

Welcome | mongosh: localhost:27017

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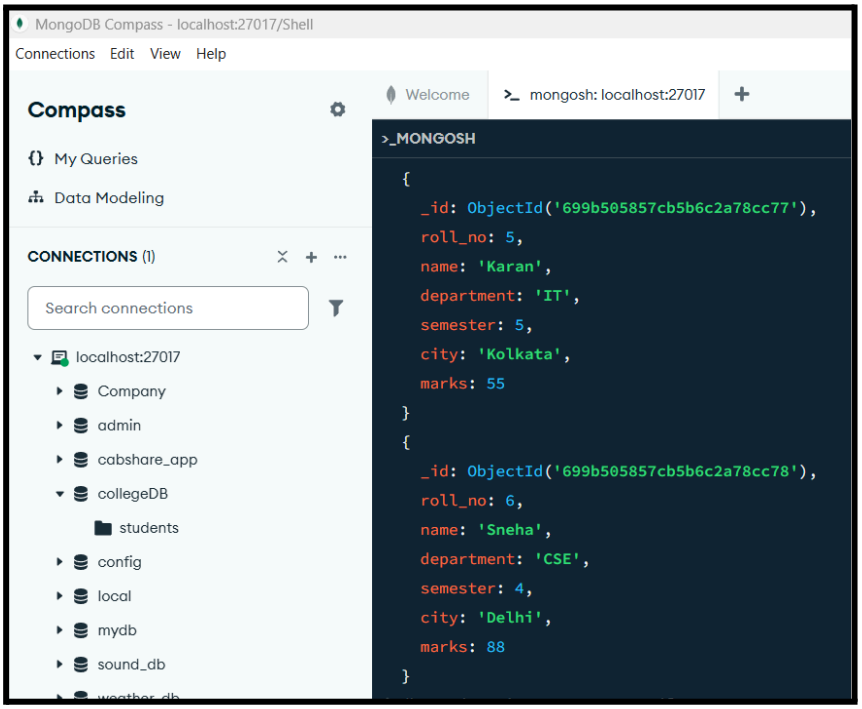
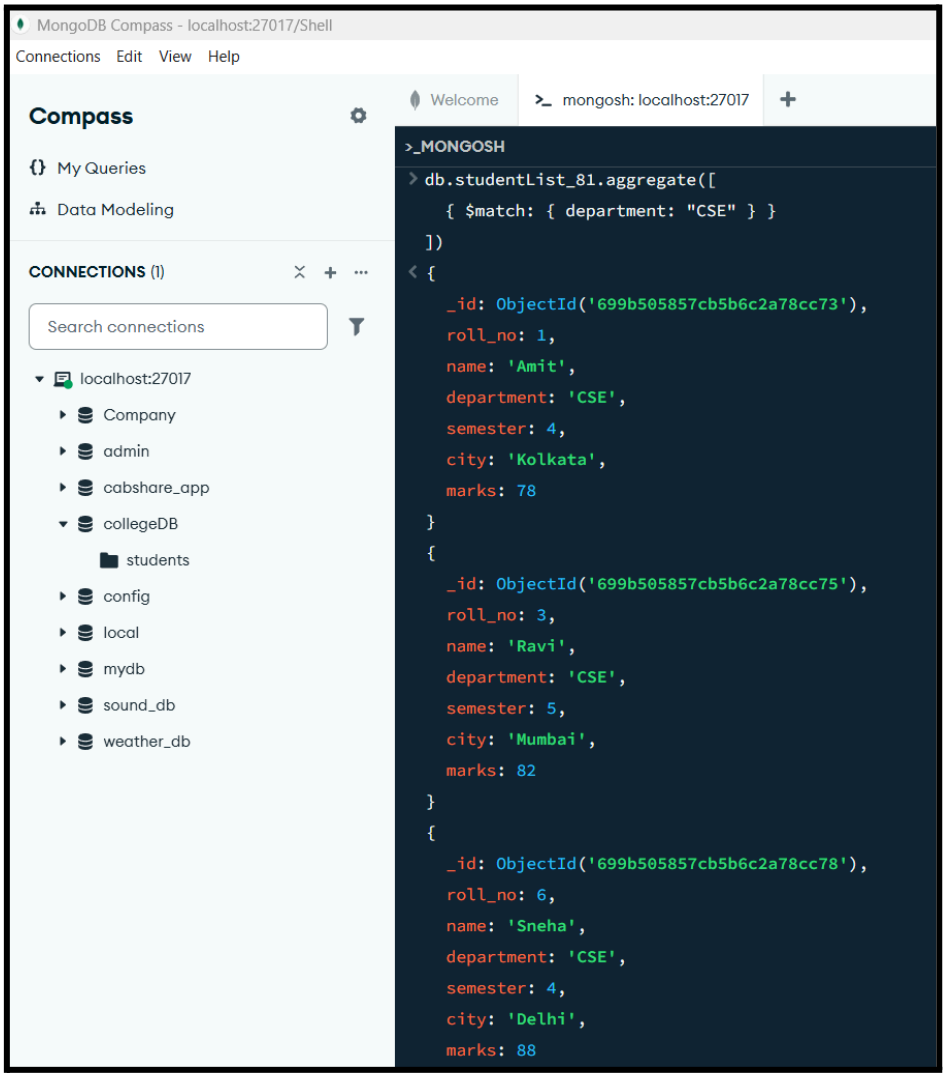
```
  {
    _id: ObjectId('699b505857cb5b6c2a78cc74'),
    roll_no: 2,
    name: 'Neha',
    department: 'IT',
    semester: 4,
    city: 'Delhi',
    marks: 65
  }
  {
    _id: ObjectId('699b505857cb5b6c2a78cc75'),
    roll_no: 3,
    name: 'Ravi',
    department: 'CSE',
    semester: 5,
    city: 'Mumbai',
    marks: 82
  }
  {
    _id: ObjectId('699b505857cb5b6c2a78cc76'),
    roll_no: 4,
    name: 'Pooja',
    department: 'ECE',
    semester: 4,
    city: 'Chennai',
    marks: 70
  }
  {
    _id: ObjectId('699b505857cb5b6c2a78cc77'),
```



Q2. Display students belonging to CSE department.



Q3. Display students of semester 4.

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cabshare_app

collegeDB

students

config

local

mydb

sound_db

weather_db

Welcome | mongosh: localhost:27017

>_MONGOSH

```
> db.studentList_81.aggregate([
    { $match: { semester: 4 } }
  ])
< {
    _id: ObjectId('699b505857cb5b6c2a78cc73'),
    roll_no: 1,
    name: 'Amit',
    department: 'CSE',
    semester: 4,
    city: 'Kolkata',
    marks: 78
  }
  {
    _id: ObjectId('699b505857cb5b6c2a78cc74'),
    roll_no: 2,
    name: 'Neha',
    department: 'IT',
    semester: 4,
    city: 'Delhi',
    marks: 65
  }
  {
    _id: ObjectId('699b505857cb5b6c2a78cc76'),
    roll_no: 4,
    name: 'Pooja',
    department: 'ECE',
    semester: 4,
    city: 'Chennai',
    marks: 70
```

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admin

cabshare_app

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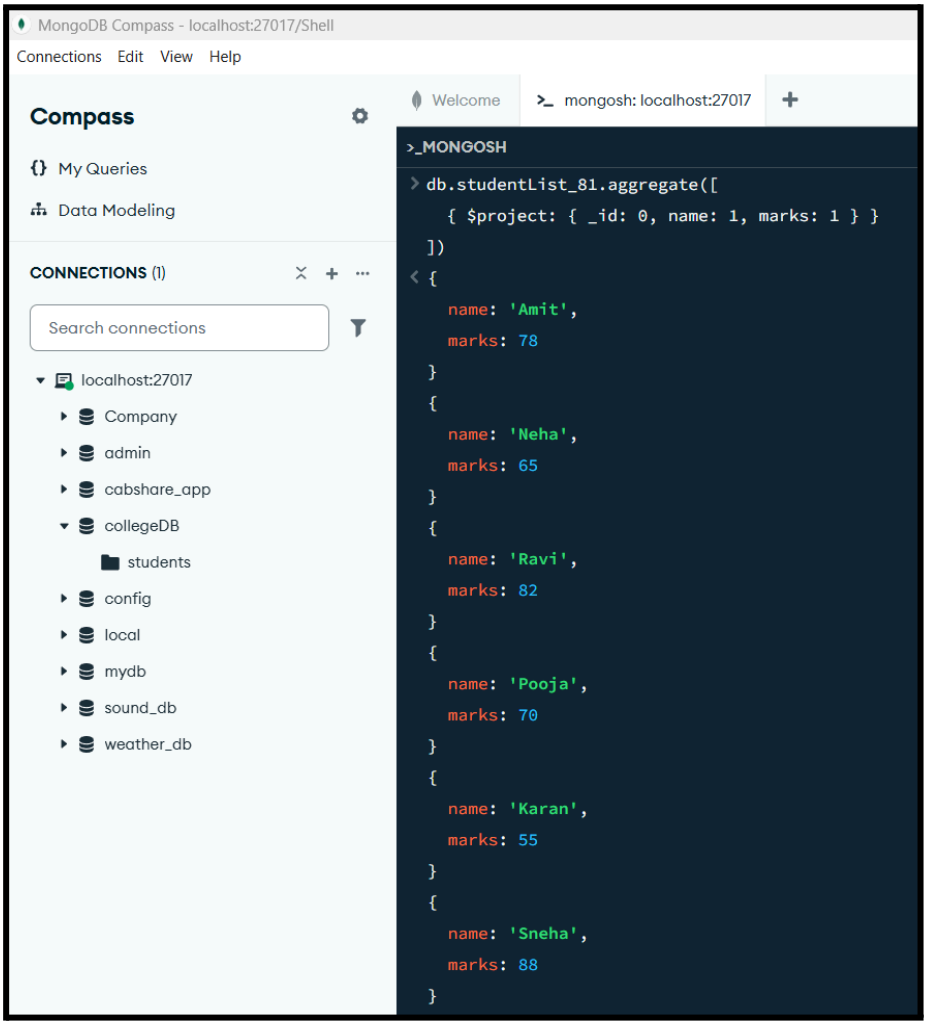
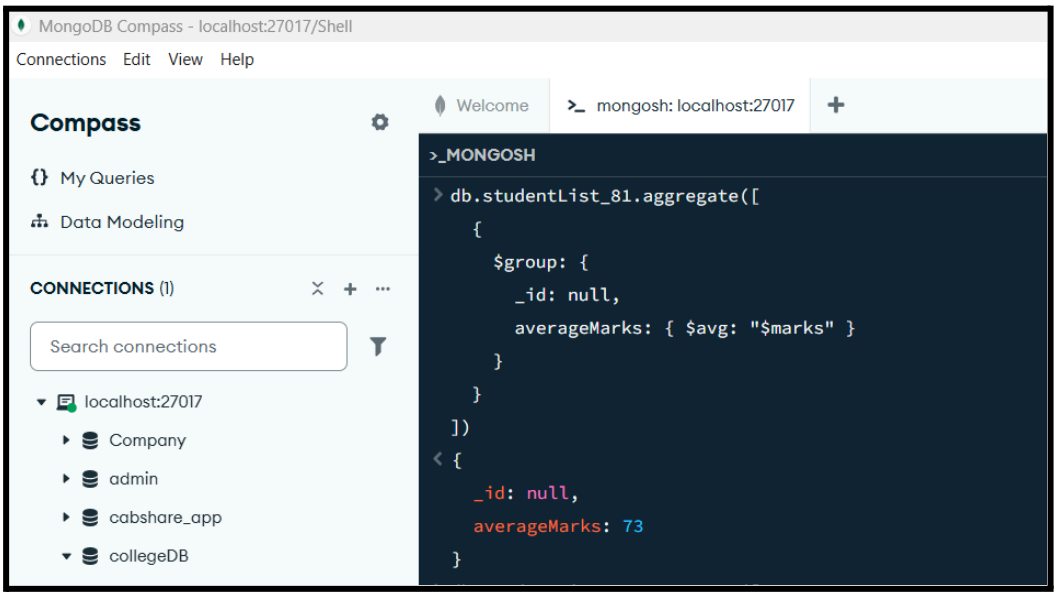
```
    marks: 70
  }
  {
    _id: ObjectId('699b505857cb5b6c2a78cc78'),
    roll_no: 6,
    name: 'Sneha',
    department: 'CSE',
    semester: 4,
    city: 'Delhi',
    marks: 88
  }
```

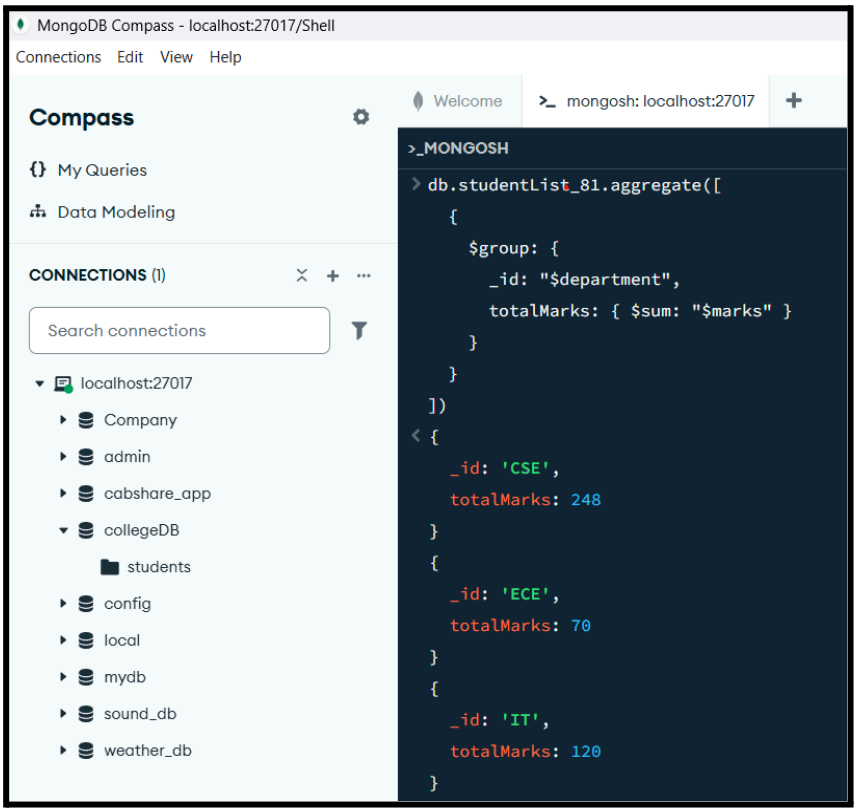
Q4. Display only name and marks of all students.



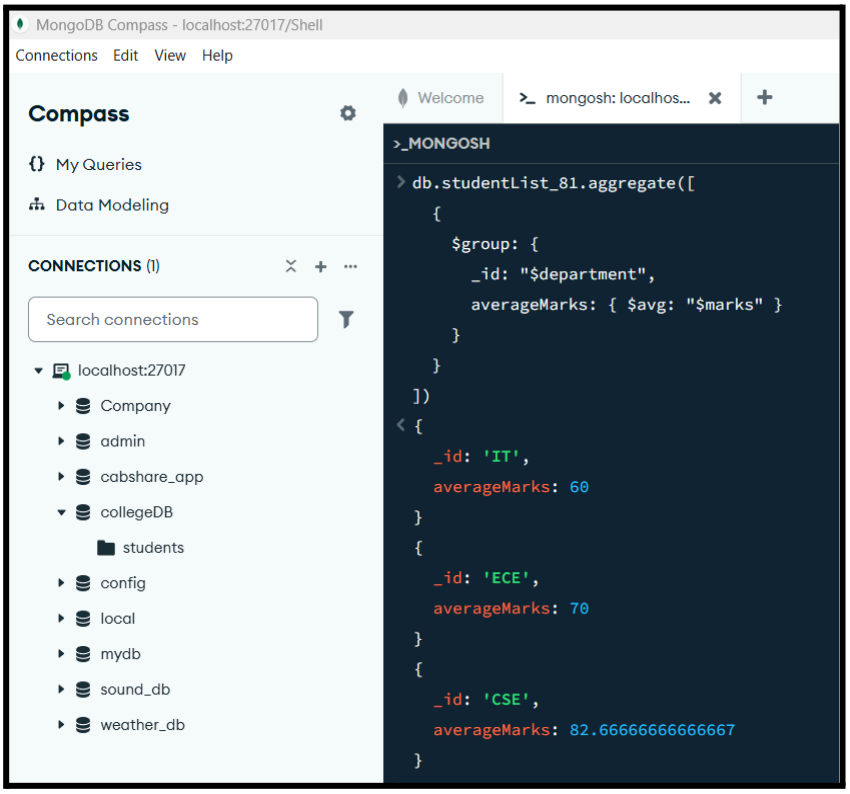
Q5. Find the average marks of all students.



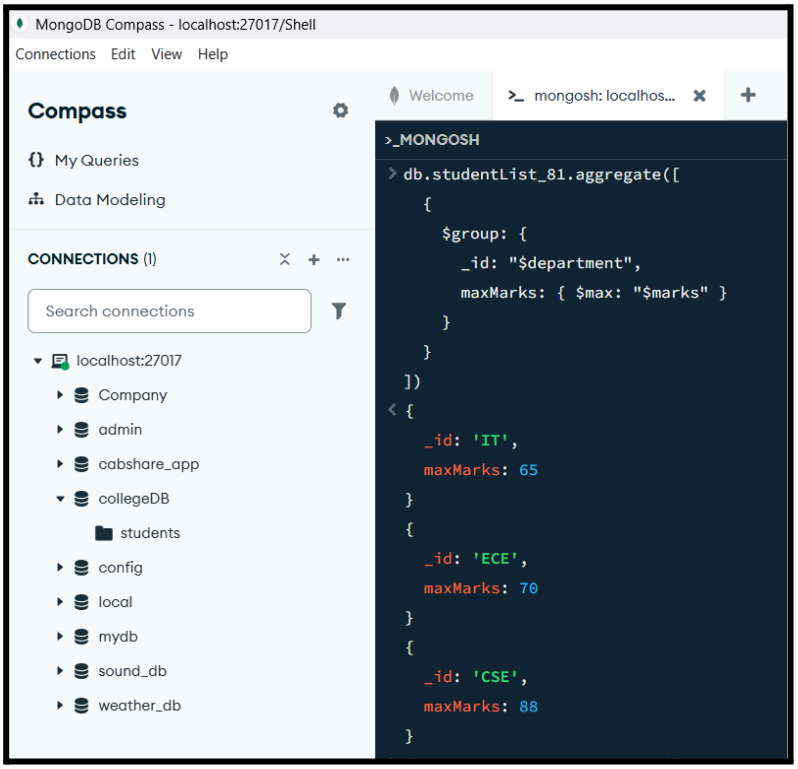
Q6. Find total marks scored department-wise.



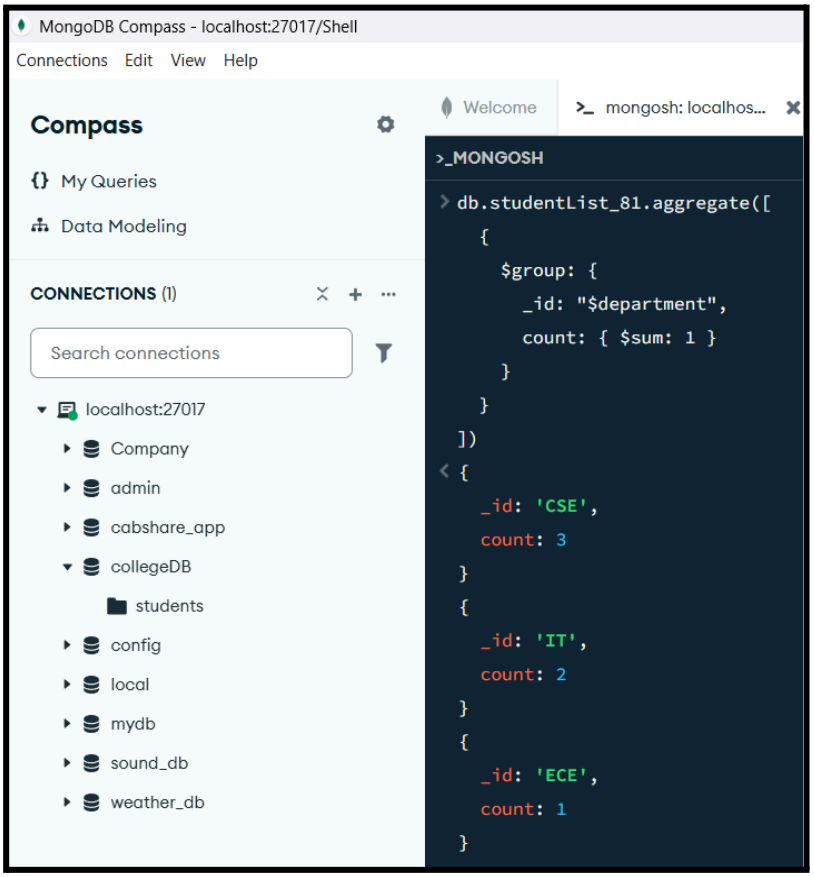
Q7. Find average marks department-wise.



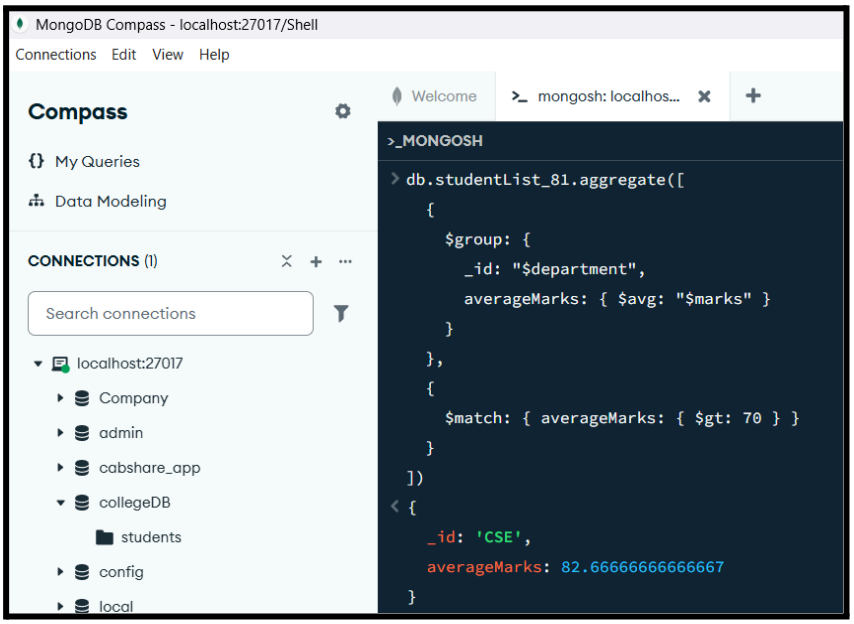
Q8. Find maximum marks scored in each department.



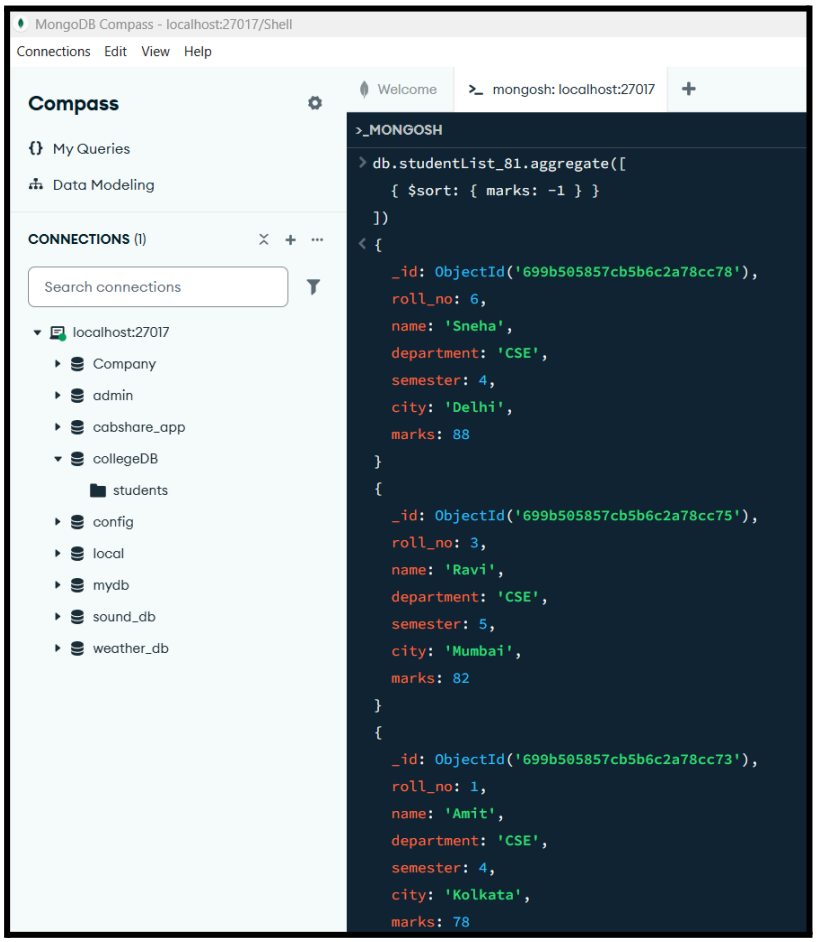
Q9. Count number of students in each department.

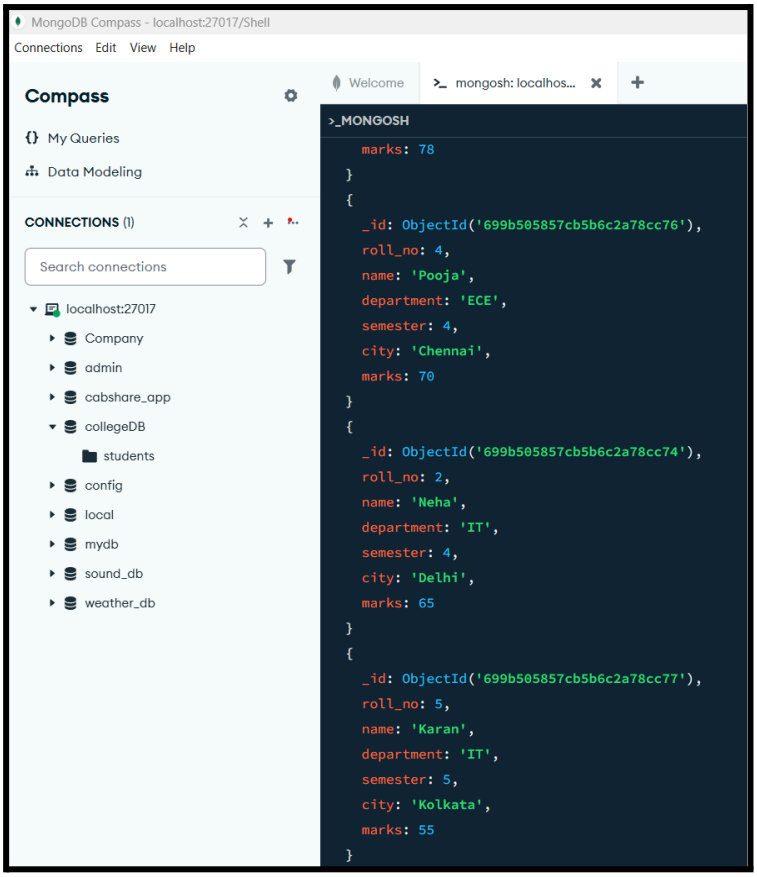


Q10. Display departments having average marks greater than 70.

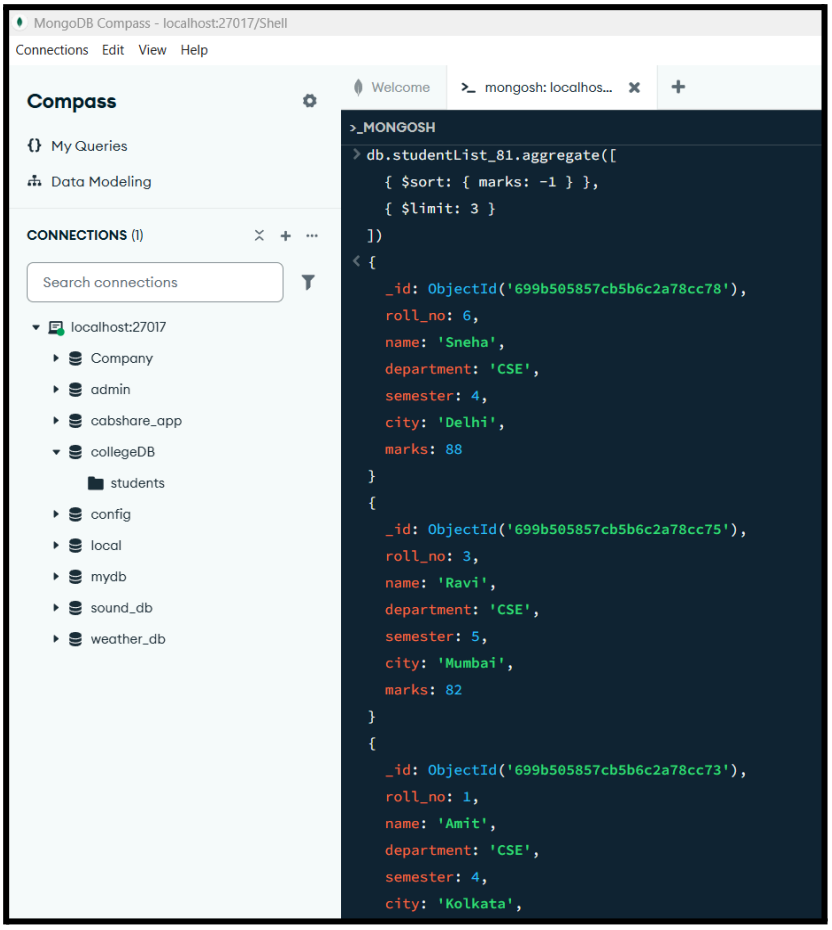


Q11. Sort students by marks in descending order.

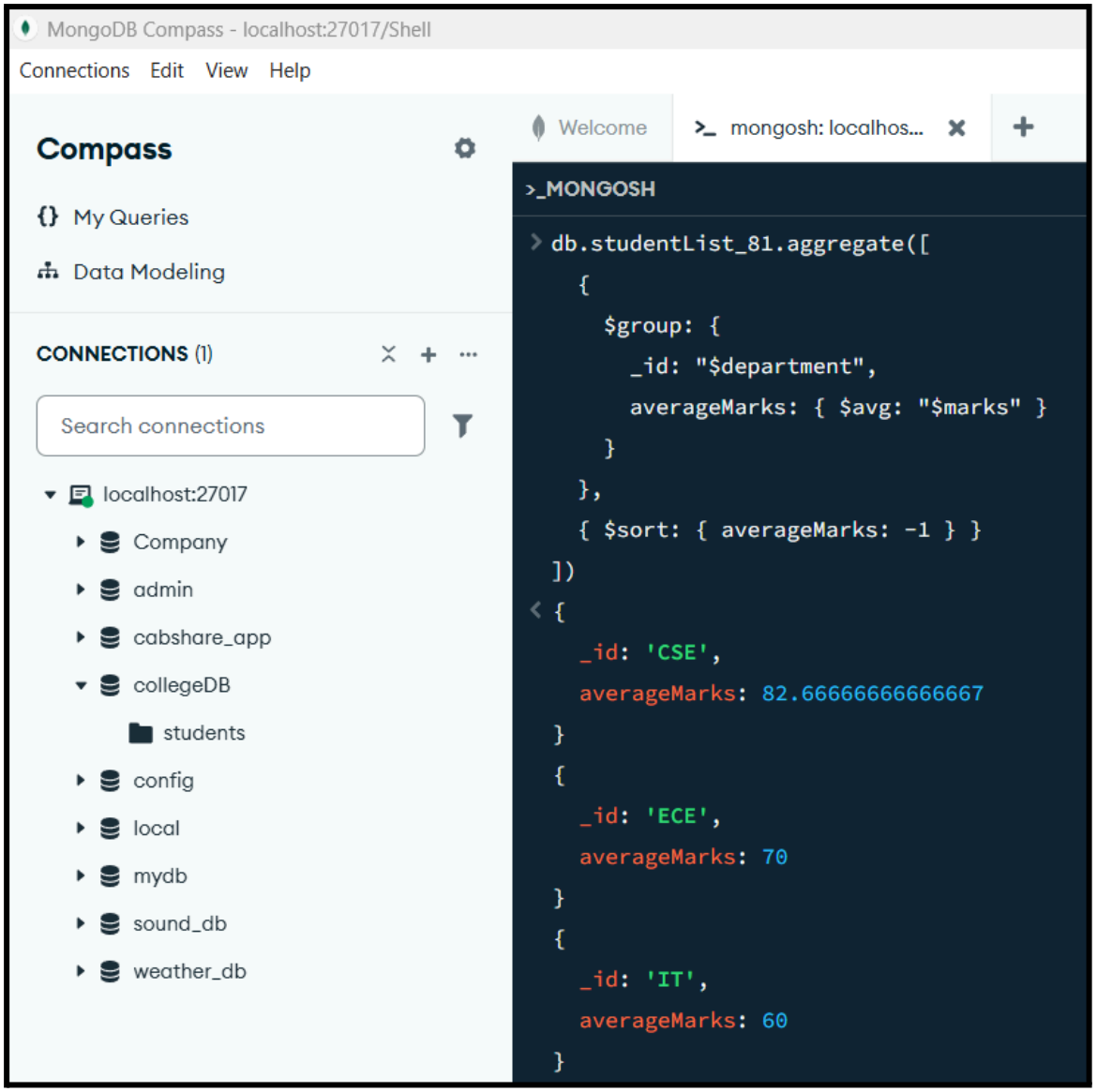
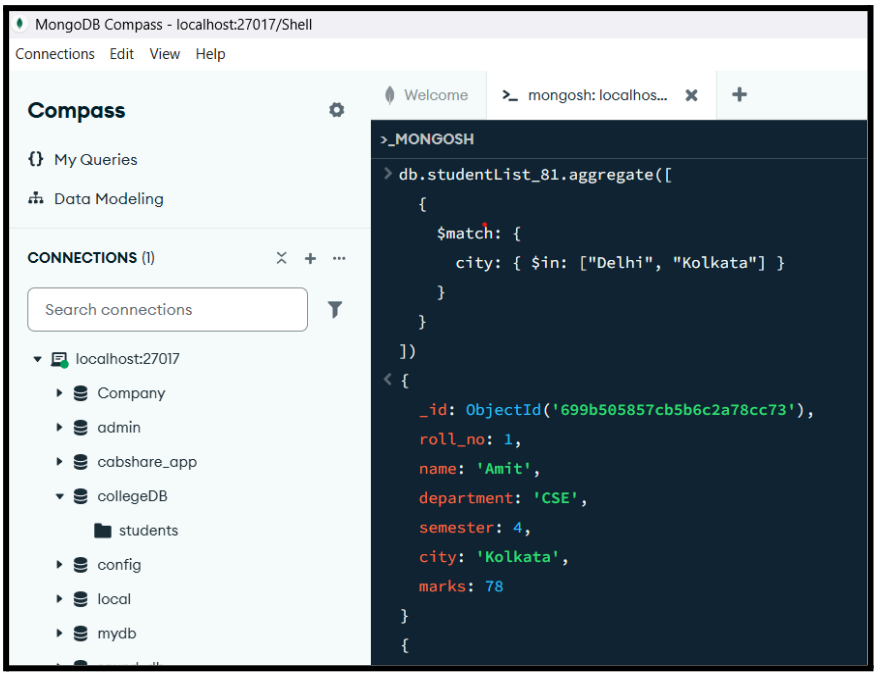
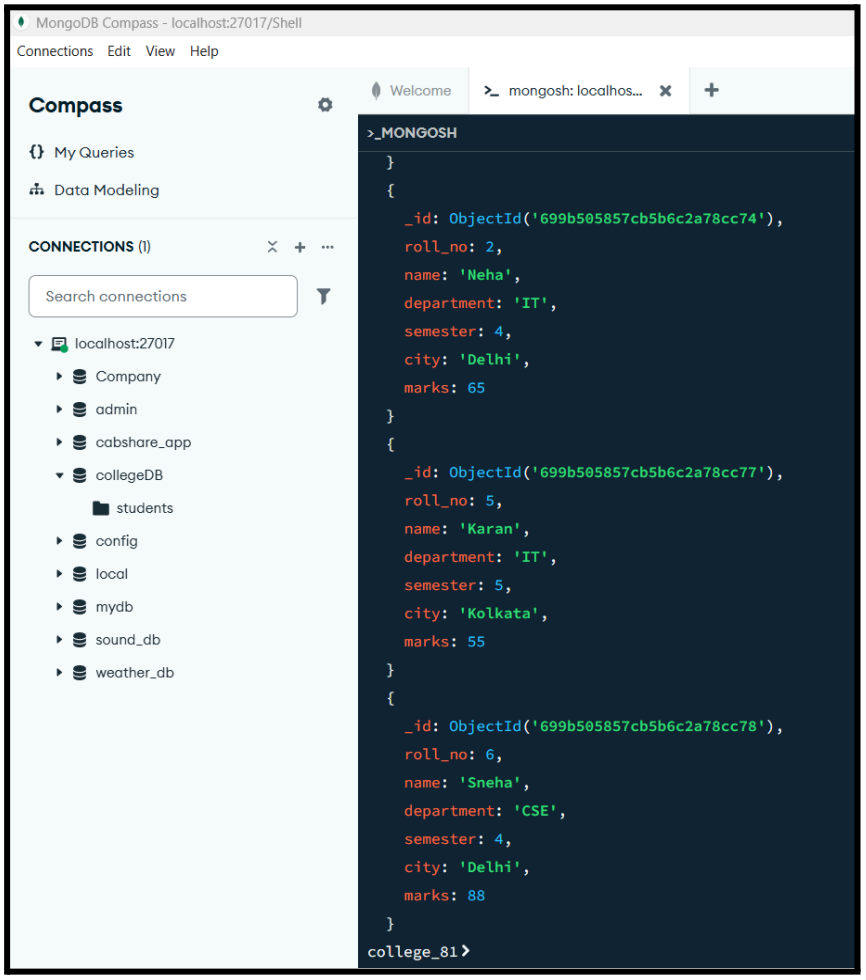




Q12. Display top 3 students based on marks.



Q13. Display department-wise average marks sorted in descending order.



Q14. Display students belonging to Delhi or Kolkata.